

Kubernetes - Deployment of an API

Prepare FastAPI App:

Create main.py file *datascientest-data-engineering/kubernetes/main.py*

Install dependencies and run

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ pip install fastapi uvicorn
Requirement already satisfied: fastapi in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (0.115.12)
Requirement already satisfied: uvicorn in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (0.33.0)
Requirement already satisfied: starlette<0.47.0,>=0.40.0 in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from fastapi) (0.44.0)
Requirement already satisfied: typing-extensions>=4.8.0 in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from fastapi) (4.13.2)
Requirement already satisfied: pydantic!=1.8,!1.8.1,!2.0.0,!2.0.1,!2.1.0,<3.0.0,>=1.7.4 in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from fastapi) (2.10.6)
Requirement already satisfied: h11>=0.8 in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from uvicorn) (0.16.0)
Requirement already satisfied: click>=7.0 in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from uvicorn) (8.1.8)
Requirement already satisfied: anyio<5,>=3.4.0 in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from starlette<0.47.0,>=0.40.0->fastapi) (4.5.2)
Requirement already satisfied: pydantic-core==2.27.2 in /home/ubuntu/datascientest-data-engineering/pyspark/.venv/lib/python3.8/site-packages (from pydantic!=1.8,!1.8.1,!2.0.0,!2.0.1,!2.1.0,<3.0.0,>=1.7.4->fastapi) (1.3.1)
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ uvicorn main:server --reload
INFO: Will watch for changes in these directories: ['/home/ubuntu/datascientest-data-engineering/kubernetes']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [10718] using WatchFiles
INFO: Started server process [10720]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Open URL <http://localhost:8000/status>

```
{
  "status": 1
}
```

Open URL <http://localhost:8000/environment>

```
{
  "environment": "unknown"
}
```

Open URL <http://localhost:8000/docs> sample execution resulted in

Curl

```
curl -X 'POST' \
  'http://localhost:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "sentence": "hello world",
    "language": "en"
  }'
```

Request URL

```
http://localhost:8000/predict
```

Server response

Code	Details
200	Response body <pre>{ "sentence": "hello world", "language": "en", "score": 0.04608502254712343 }</pre> Response headers <pre>content-length: 70 content-type: application/json date: Wed, 02 Jul 2025 17:18:38 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

The Docker Image:

Pull image from Docker Hub

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ docker pull datascientest/fake-api:1.0.0
1.0.0: Pulling from datascientest/fake-api
540db60ca938: Pull complete
176c2259dfae: Pull complete
8acb92f0f933: Pull complete
Digest: sha256:1e6f513eef068753314191ea5674b9bbd671487dfe7158bc8012da0003082d75
Status: Downloaded newer image for datascientest/fake-api:1.0.0
docker.io/datascientest/fake-api:1.0.0
```

Verify its presence locally

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ docker images datascientest/fake-api
REPOSITORY          TAG          IMAGE ID          CREATED          SIZE
datascientest/fake-api 1.0.0       437c186f35ae     4 years ago     72MB
```

Run it locally to confirm it serves FastAPI app

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ docker run --rm -p 8000:8000 datascientest/fake-api:1.0.0
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

In another terminal run the following commands

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ curl http://localhost:8000/status
{"status":1}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ curl http://localhost:8000/environment
{"environment":"unknown"}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ curl -X POST http://localhost:8000/predict \
> -H "Content-Type: application/json" \
> -d '{"sentence":"test","language":"en"}'
{"sentence":"test","language":"en","score":0.5768446838198626}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Pod Deployment:

Create my-api.yml file *datascientest-data-engineering/kubernetes/my-api.yml*

```
! my-api.yml U X
datascientest-data-engineering > kubernetes > ! my-api.yml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: my-sentiment-analysis-deployment
5    labels:
6      app: my-sentiment-analysis-api
7  spec:
8    replicas: 3
9    selector:
10     matchLabels:
11       app: my-sentiment-analysis-api
12   template:
13     metadata:
14       labels:
15         app: my-sentiment-analysis-api
16     spec:
17       containers:
18         - name: my-sentiment-analysis-api
19           image: datascientest/fake-api:1.0.0
20           ports:
21             - containerPort: 8000
```

Apply it

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f my-api.yml
deployment.apps/my-sentiment-analysis-deployment created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get deploy
NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
my-sentiment-analysis-deployment    0/3      3              0             6s
nginx-deployment                    1/1      1              1            32h
vote                                12/12    12             12           7h21m
```

API exposure:

Create my-service.yml file

```
! my-service.yml U X
datascientest-data-engineering > kubernetes > ! r
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: my-sentiment-analysis-service
5    labels:
6      app: my-sentiment-analysis-api
7  spec:
8    type: ClusterIP
9    ports:
10     - port: 8001
11       targetPort: 8000
12       protocol: TCP
13    selector:
14      app: my-sentiment-analysis-api
```

Apply it

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f my-service.yml
service/my-sentiment-analysis-service created
```

Create my-ingress.yml file

```
! my-ingress.yml U X
datascientest-data-engineering > kubernetes > ! my-in
1  apiVersion: networking.k8s.io/v1
2  kind: Ingress
3  metadata:
4    name: my-sentiment-analysis-ingress
5  spec:
6    defaultBackend:
7      service:
8        name: my-sentiment-analysis-service
9        port:
10         number: 8001
```

Apply it

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f my-ingress.yml
ingress.networking.k8s.io/my-sentiment-analysis-ingress unchanged
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get ingress
NAME                                CLASS    HOSTS    ADDRESS    PORTS    AGE
my-sentiment-analysis-ingress      traefik  *       80         19m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get ingress my-sentiment-analysis-ingress
NAME                                CLASS    HOSTS    ADDRESS    PORTS    AGE
my-sentiment-analysis-ingress      traefik  *       80         20m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get svc --all-namespaces | grep traefik
kube-system      traefik                                LoadBalancer    10.43.193.11    176.34.128.201    80:31828/TCP,443:30570/TCP    2d9h
```

Port-forward the service locally

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl port-forward svc/my-sentiment-analysis-service 8000:8001
Forwarding from 127.0.0.1:8000 -> 8000
Forwarding from [::1]:8000 -> 8000
```

Point to browser <http://localhost:8000/docs>

My API 0.1.0 OAS3
/openapi.json

default

GET	/status	Get Status	▼
GET	/environment	Get Environment	▼
POST	/predict	Post Sentence	▼

Schemas

HTTPValidationError >

PredictedSentence >

Sentence >

ValidationError >

Adding a ConfigMap:

Create a ConfigMap

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl create configmap my-config-map --from-literal ENVIRONMENT_TYPE=production
configmap/my-config-map created
```

Create my-api2.yml file

```
my-api2.yml U X
tascientest-data-engineering > kubernetes > ! my-api2
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: my-sentiment-analysis-deployment
5    labels:
6      app: my-sentiment-analysis-api
7  spec:
8    replicas: 3
9    selector:
10     matchLabels:
11       app: my-sentiment-analysis-api
12    template:
13     metadata:
14       labels:
15         app: my-sentiment-analysis-api
16     spec:
17       containers:
18       - name: my-sentiment-analysis-api
19         image: datascientest/fake-api:1.0.0
20         ports:
21         - containerPort: 8000
22         envFrom:
23         - configMapRef:
24           name: my-config-map
```

Delete my-sentiment-analysis-deployment

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl delete deployment my-sentiment-analysis-deployment
deployment.apps "my-sentiment-analysis-deployment" deleted
```

Apply

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f my-api2.yml
deployment.apps/my-sentiment-analysis-deployment created
```

Await roll out

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl rollout status deployment/my-sentiment-analysis-deployment
deployment "my-sentiment-analysis-deployment" successfully rolled out
```

Re-establish port

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl port-forward svc/my-sentiment-analysis-service 8000:8001
Forwarding from 127.0.0.1:8000 -> 8000
Forwarding from [::1]:8000 -> 8000
```

Test /environment endpoint locally

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ curl http://127.0.0.1:8000/environment
{"environment":"production"}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

```
{
  "environment": "production"
}
```

Adding a Secret:

Create the secret

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl create secret generic my-secret --from-literal my-key=my-value
secret/my-secret created
```

Create my-api3.yml file

```
my-api3.yml U X
datascientest-data-engineering > kubernetes > ! my-api3
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: my-sentiment-analysis-deployment
5    labels:
6      app: my-sentiment-analysis-api
7  spec:
8    replicas: 3
9    selector:
10     matchLabels:
11       app: my-sentiment-analysis-api
12   template:
13     metadata:
14       labels:
15         app: my-sentiment-analysis-api
16     spec:
17       containers:
18       - name: my-sentiment-analysis-api
19         image: datascientest/fake-api:1.0.0
20         ports:
21         - containerPort: 8000
22         env:
23         - name: ENVIRONMENT_TYPE
24           valueFrom:
25             secretKeyRef:
26               name: my-secret
27               key: my-key
```

Remove old deployment

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl delete deployment my-sentiment-analysis-deployment
deployment.apps "my-sentiment-analysis-deployment" deleted
```

Apply new deployment

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f my-api3.yml
deployment.apps/my-sentiment-analysis-deployment created
```

Await roll out finish

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl rollout status deployment/my-sentiment-analysis-deployment
deployment "my-sentiment-analysis-deployment" successfully rolled out
```

Re-establish port-forwarding

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl port-forward svc/my-sentiment-analysis-service 8000:8001
Forwarding from 127.0.0.1:8000 -> 8000
Forwarding from [::1]:8000 -> 8000
```

Verify the secret was injected

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ curl http://127.0.0.1:8000/environment
{"environment":"my-value"}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

```
{
  "environment": "production"
}
```